

Assignment 1

n firms $I = \{1, \dots, n\}$ are extracting a limited resource with q_i being how much they extract each

a firm's profits can be calculated

$$\begin{aligned}\pi_i &= q_i \left[1 - \left(\sum_{s \in I} q_s \right)^2 \right] = q_i - q_i \cdot \left(q_i + \sum_{s \in I \setminus i} q_s \right)^2 \\ &= q_i - q_i \cdot (q_i^2 + 2q_i \cdot \sum_{s \in I \setminus i} q_s + \left(\sum_{s \in I \setminus i} q_s \right)^2) = q_i - q_i^3 - 2q_i^2 \cdot \sum_{s \in I \setminus i} q_s - q_i \cdot \left(\sum_{s \in I \setminus i} q_s \right)^2\end{aligned}$$

If $\sum_{s=1}^n q_s \leq 1$
0 otherwise

We find the best response function for firm i

Since it simply want to maximize profits we take the FOC by using the product rule and the chain rule

We set

$$\begin{aligned}y &= \sum_{s \in I \setminus i} q_s \\ \frac{\partial \pi_i}{\partial q_i} &= 1 - 3q_i^2 - 4q_i \cdot y - y^2 = 0\end{aligned}$$

We call the sum of the other q 's y and solve the equation

$$1 - 3q_i^2 - 4q_i \cdot y - y^2 = 0$$

$\Updownarrow$

$$q_i = \frac{\sqrt{y^2 + 3} - 2 \cdot y}{3} \quad \vee \quad q_i = -\left(\frac{\sqrt{y^2 + 3} + 2 \cdot y}{3} \right)$$

We get 2 solutions but notice that the one is never valid since we only extract positive amounts (since otherwise we get negative profits which we would never want since $q_i = 0$ is a valid solution)

Thus

$$\frac{\sqrt{y^2 + 3} - 2 \cdot y}{3}$$

And

$$B_i(q) = \begin{cases} \frac{\sqrt{y^2 + 3} - 2y}{3} & \text{for } y < 1 \\ 0 & \text{ellers} \end{cases}$$

By symmetry in EQ we must have

$$q_1 = q_2 = \dots = q_n$$

If $y=1$ then all profits are 0 then this can't be Nash since by extracting less someone will get higher profits

By that argument we can say that

$$y < 1$$

Thus

We notice that

$$\frac{\sqrt{y^2 + 3} - 2y}{3}$$

Thus symmetry implies

$$q_i = \frac{\sqrt{((n-1) \cdot q_i)^2 + 3} - 2(n-1)q_i}{3}$$

$\Updownarrow$

$$q_i = -\left(\frac{1}{\sqrt{n^2 + 2 \cdot n}}\right) \vee q_i = \frac{1}{\sqrt{n^2 + 2 \cdot n}}$$

Again negative q_i doesn't make sense because it implies negative profit which can never be optimal then the solution must be

$$q_i = \frac{1}{\sqrt{n^2 + 2 \cdot n}}$$

This gives every firm the profits

$$\begin{aligned} \pi_i &= \frac{1}{\sqrt{n^2 + 2 \cdot n}} \cdot \left(1 - \left(\frac{n}{\sqrt{n^2 + 2 \cdot n}}\right)^2\right) = \frac{1}{\sqrt{n^2 + 2 \cdot n}} \left(1 - \frac{n^2}{n^2 + 2n}\right) \\ &= \frac{1}{\sqrt{n^2 + 2 \cdot n}} \cdot \left(\frac{2n}{n^2 + 2n}\right) \end{aligned}$$

b)

maximizing the sum of profits

we call $\sum_{i \in I} q_i = Q$

$$\Pi = \sum_{i \in I} \pi_i = Q \cdot (1 - Q^2)$$

We set FOC

$$\frac{\partial \Pi}{\partial Q} = 1 - 3Q^2 = 0$$

$$Q^* = \frac{1}{\sqrt{3}}$$

$$\Pi^* = \frac{1}{\sqrt{3}} \cdot \left(1 - \frac{1}{3}\right) = \frac{2}{\sqrt{3} \cdot 3}$$

This is of course also a special case of (a) with $n=1$ as the actors are acting as a unit to maximize the sum of their profits

Assuming an even allocation of profits

Every actor will then get

$$\pi_i = \frac{2}{\sqrt{3} \cdot 3 \cdot n}$$

We note that this profit is larger than when they are competing because

$$\frac{2}{\sqrt{3} \cdot 3 \cdot n} \geq \frac{1}{\sqrt{n^2 + 2 \cdot n}} \cdot \left(\frac{2n}{n^2 + 2n} \right) \Leftrightarrow$$

$$\Leftrightarrow \frac{4}{3 \cdot 9 \cdot n^2} \geq \frac{1}{n^2 + 2n} \cdot \frac{4n}{(n^2 + 2n)^2} \Leftrightarrow \frac{1}{3 \cdot 9} \geq \frac{1}{(n + 2)^3} \Leftrightarrow (n + 2)^3 \geq 3^3 \text{ for } n \geq 1$$

c) the situation is very much like the oligopol duopoly model we could see it as a special of exactly that model with

$$P(Q) = \begin{cases} 1 - Q^2 & \text{for } Q < 1 \\ 0 & \text{ellers} \end{cases}$$

And no costs

We also see the clear relation between competing and cooperating where cooperating gives you higher profits just like when firms make a cartel in the oligopoly model and thus get higher profits but will want to deviate unless the game is repeated and some sort of consequence is reasoned